CPU Scheduling — SRTF & Priority Scheduling

SRTF
Shortest Remaining Time First

Definition
SRTF is the preemptive version of SJF.
The scheduler always selects the process with the:
Shortest Remaining CPU Burst Time

SJF vs SRTF

Feature	SJF	SRTF
Type	Non-preemptive	Preemptive
Selection	Shortest burst	Shortest remaining time
Current process can be interrupted?	No	Yes
New process can cause preemption?	No	Yes
Gantt chart	Simpler	More complex
Context switches	Generally fewer	Generally more

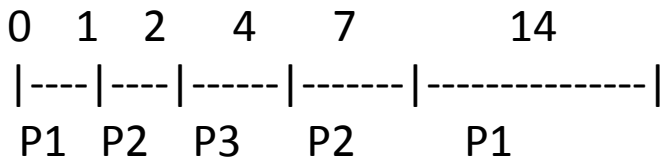
SJF = Shortest Job
SRTF = Shortest Remaining Time

SRTF Basic Example

Process	AT	BT
P1	0	8
P2	1	4

SRTF Numerical 1

Process	AT	BT
P1	0	8
P2	1	4
P3	2	2



P	AT	BT	CT	TAT	WT
P1	0	8	14	14	6
P2	1	4	7	6	2
P3	2	2	4	2	0

Average WT

$$AWT = \frac{6 + 2 + 0}{3} \boxed{AWT = 2.67}$$

EX 2

Process	AT	BT
P1	0	7
P2	2	4
P3	3	1
P4	5	2

EX 2

Process	AT	BT	CT	TAT	WT
P1	0	7	14	14	7
P2	2	4	7	5	1
P3	3	1	4	1	0
P4	5	2	9	4	2

Average waiting time = 2.500000
Average Turnaround time = 6.000000

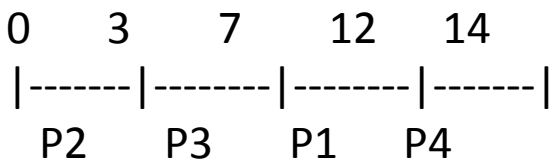
Priority Scheduling

Process	Priority
P1	3
P2	1
P3	4
P4	2

Non-Preemptive Priority

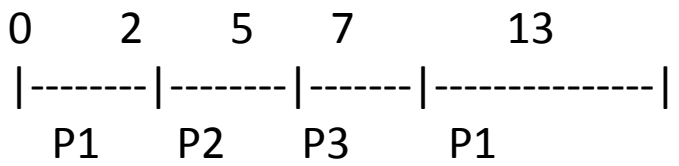
Process	AT	BT	Priority
P1	0	5	3
P2	0	3	1
P3	0	4	2
P4	0	2	4

P	AT	BT	Pri	CT	TAT	WT
P1	0	5	3	12	12	7
P2	0	3	1	3	3	0
P3	0	4	2	7	7	3
P4	0	2	4	14	14	12



Preemptive Priority Scheduling

Process	AT	BT	Priority
P1	0	8	3
P2	2	3	1
P3	3	2	2


0
2
5
7
13
P1
P2
P3
P1

EX

Process	AT	BT	Priority
P1	0	6	3
P2	1	4	1
P3	2	2	4
P4	3	5	2

Process	AT	BT	Priority	CT
P1	0	6	3	15
P2	1	4	1	5
P3	2	2	4	17
P4	3	5	2	10

